WNUT17 NER, Part 2 Section B: Pooled 80/10/10 Split

Kelompok 1, Pemrosesan Bahasa Alami Lanjut. Addendum to the Section A report (official WNUT17 split). This experiment pools the three official files and re-splits them at random. Mixing was authorized for this section only. The result below is an ablation, not a new leaderboard number.

1. What Section B does and why

Section A keeps the official WNUT17 design: train is Twitter 2010, dev is YouTube, test is Reddit, StackExchange and Twitter 2017. That separation is deliberate. It forces every model to tag entities it has never seen, which is the whole difficulty of the task. Section B asks the opposite question: if we ignore the design and shuffle everything into a conventional random split, how much easier does the task get, and why. We build the pooled split, run the same three models with the same finalized hyperparameters, and read the gap against Section A.

2. Building the split

The split unit is the document, keyed as (source file, document id) so a tweet's sequences never cross splits and no document from one original file collides with another. We shuffle the 3296 documents with numpy RandomState(42) and cut 80/10/10 by document count (2636/329/331 documents). Three new CoNLL files go to data_partB/.

Split	Sequences	Tokens	Entities	Seq %	Token %	Ent %
Train	5030	90785	3378	88.4	89.13	86.84
Dev	329	5434	256	5.78	5.33	6.58
Test	331	5638	256	5.82	5.54	6.58
Total	5690	101857	3890	100.0	100.0	100.0

Conservation check passes exactly: the three new files hold 5,690 sequences, 101,857 tokens and 3,890 entities, identical to the pooled originals. BIO violations are zero in all three files. Because documents vary in size, the achieved entity percentages deviate from 10 percent for dev and test; this is measured, not assumed, and is reported above.

3. Entity leakage, the main driver

Before training anything, we measure how many gold entity mentions in the new dev and test also appear as an entity in the new train. On the official split this figure is about 2 percent for test against train, by design. On the pooled split it rises to 18.4 percent for the new dev and 21.1 percent for the new test. 81 distinct entity surfaces now appear in at least two splits, led by frequent names such as trump, god, chicago, london, ireland, obama. This leakage number sits next to every F1 in this section, because it is the main reason the numbers move.

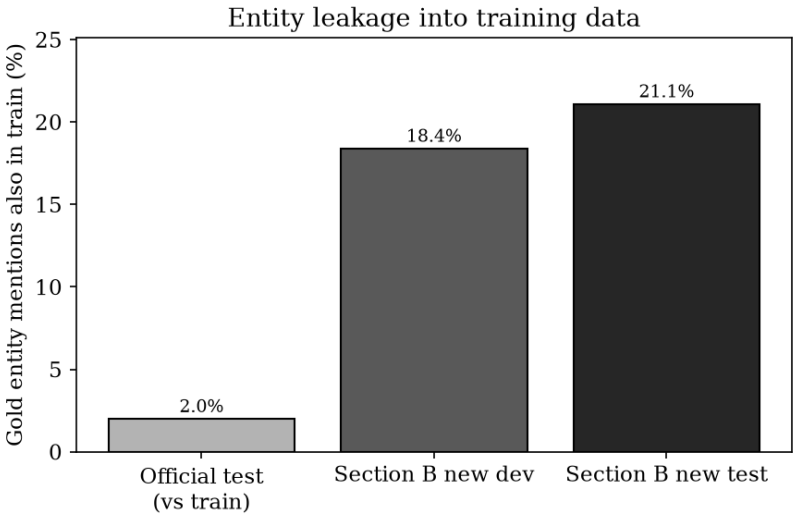


Figure 1. Entity leakage into the training data: official test vs the pooled split.

4. Results: Section A vs Section B

Model	Split	P	R	F1 entity	F1 surface	Test leakage
CRF	A official	31.9	11.4	16.8	17.6	~2%
CRF	B pooled	54.2	32.4	40.6	38.5	21.1%
SVM	A official	31.5	28.3	29.8	26.0	~2%
SVM	B pooled	36.8	49.2	42.1	35.5	21.1%
Bi-LSTM CRF	A official	57.4	31.1	40.4	39.3	~2%
Bi-LSTM CRF	B pooled	58.0	46.9	51.8	50.8	21.1%

All Section B test scores carry a 2000-resample bootstrap interval: CRF 40.5 (95% CI 33.9 to 47.3); SVM 36.0 (95% CI 30.8 to 41.2); Bi-LSTM CRF 51.8 (95% CI 45.4 to 58.4).

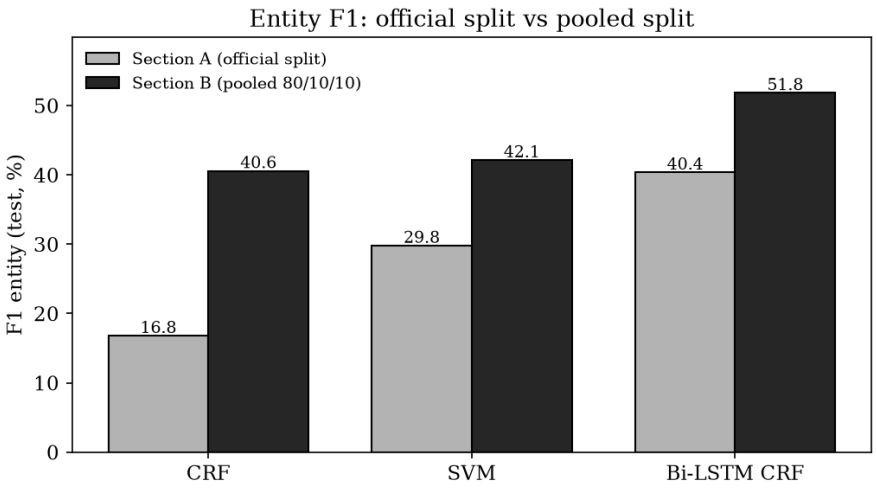


Figure 2. Entity F1 on the official split (Section A) and the pooled split (Section B).

5. Why the numbers move, and what they do not mean

The pooled split is easier for two separate reasons, and they must not be confused with a better model. First, leakage: entities that recur across the corpus now show up in both train and test, so the model can memorize them. Second, feature coverage: with a random 80 percent of all documents in train, the CRF and SVM simply see more vocabulary, more word shapes and more context patterns than the Twitter-only train file offered. Neither effect is available on the official split, and neither reflects better handling of novel entities.

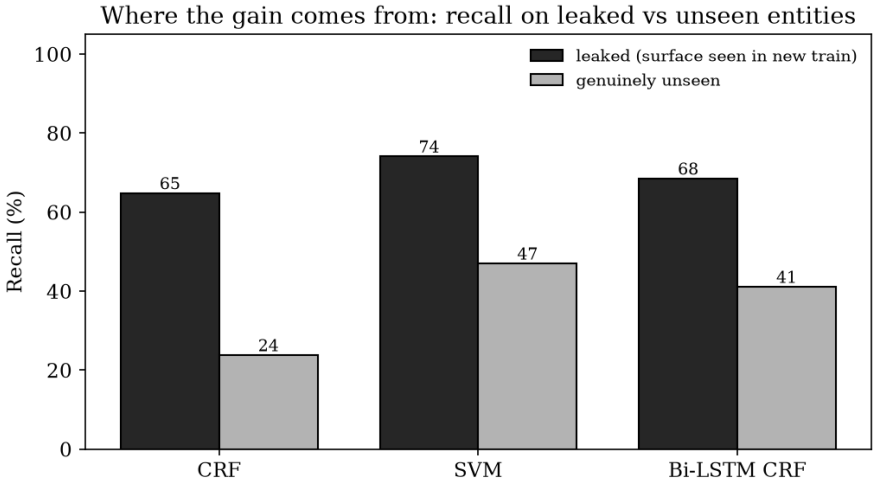


Figure 3. Recall split by whether the test entity leaked into the new train or is still genuinely unseen. The gain concentrates on leaked entities; recall on genuinely unseen entities stays close to the Section A regime.

This is why the Section B result is not comparable to the WNUT17 shared task. The leaderboard, and the 41.86 shared-task best, are defined on the domain-separated split whose premise is that test entities are novel. Reshuffling removes that premise. Section B should be read as an ablation that isolates how much of the task's difficulty comes from novel and rare entities specifically, rather than from general NER difficulty. On this evidence, most of the difficulty is the novelty: remove it and even the lexical CRF, which scored 16.8 on the official split, moves into the 40s.

6. Per-type, confusion and errors

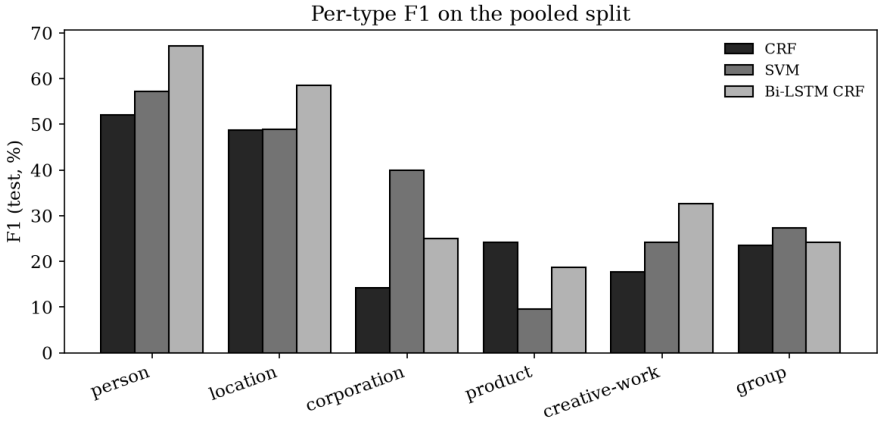


Figure 4. Per-type F1 on the pooled split.

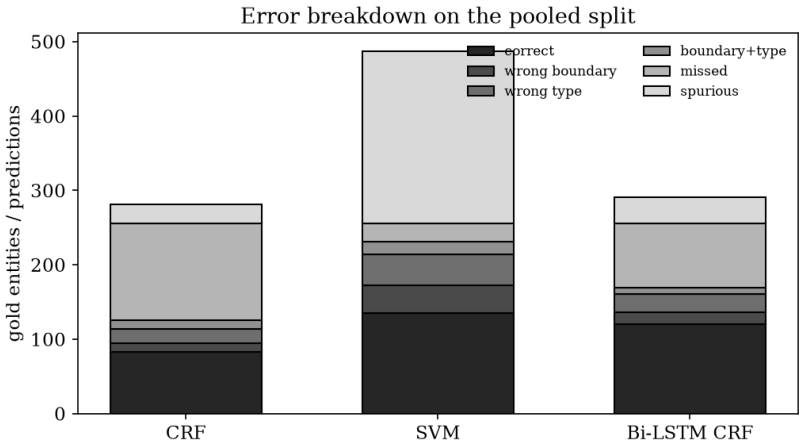


Figure 5. Error breakdown on the pooled split.

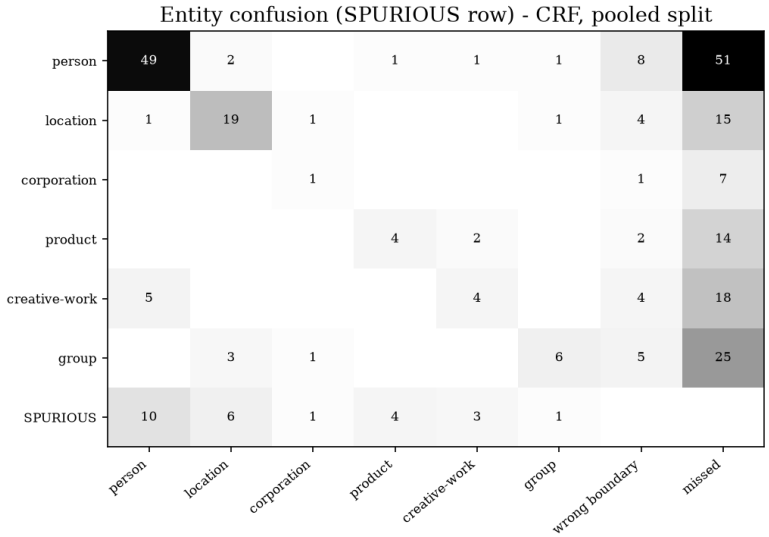


Figure. Entity confusion with a SPURIOUS row, CRF.

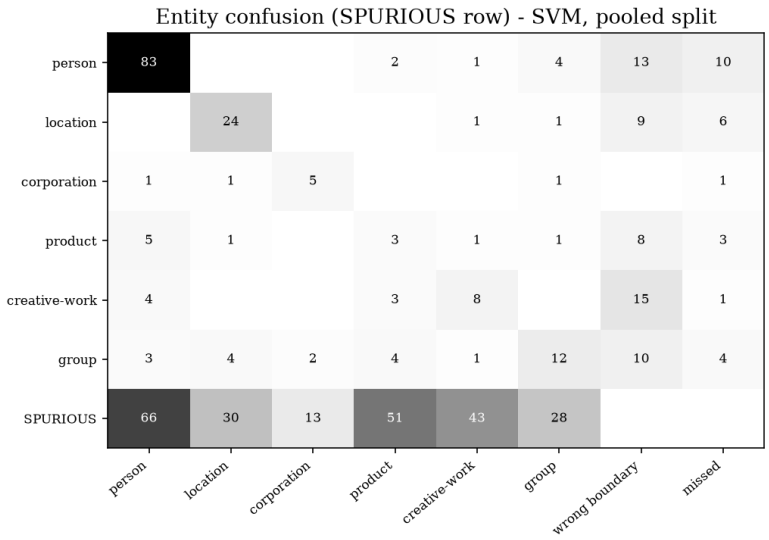


Figure. Entity confusion with a SPURIOUS row, SVM.

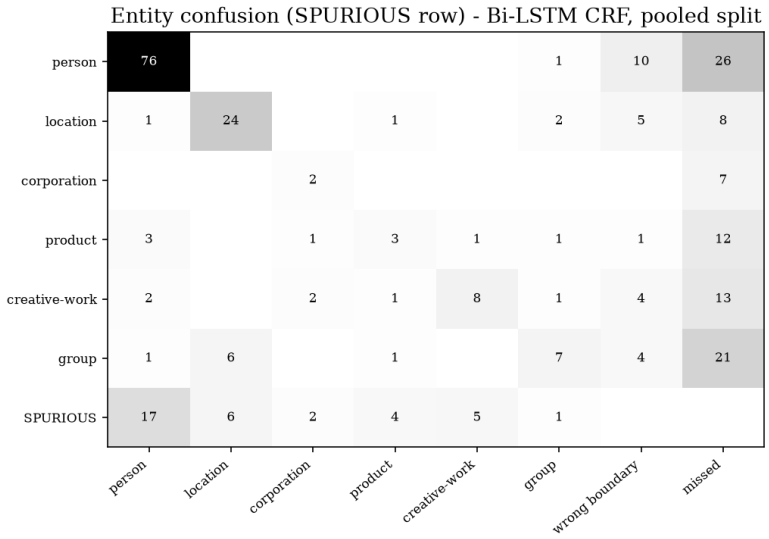


Figure. Entity confusion with a SPURIOUS row, Bi-LSTM CRF.

7. Conclusion

On the pooled 80/10/10 split every model improves sharply, with Bi-LSTM CRF reaching F1 51.8 on the new test. The improvement tracks a leakage rate of 21.1 percent, up from about 2 percent on the official split, and extra feature coverage for the classical models. It is an ablation of the dataset's difficulty, not a better NER system, and it does not compare to the shared-task leaderboard. Section A remains the primary, honest result for this assignment.